

OPERATING SYSTEM LAB FILE ASSIGNMENT-2

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FCFS and SJF Scheduling

1. FCFS CODE

```
2 # -----
3 # FCFS Scheduling
4 # -----
5
6 def fcfs(processes):
7     current_time = 0
8     intervals = []
9
10    # FCFS: arrange according to arrival time
11    ordered = sorted(
12        processes,
13        key=lambda p: (p["arrival"], p["pid"])
14    )
15
16    for process in ordered:
17
18        # CPU idle if process has not arrived
19        if current_time < process["arrival"]:
20            intervals.append(
21                ("IDLE", current_time, process["arrival"])
22            )
23            current_time = process["arrival"]
24
25        start = current_time
26        end = start + process["burst"]
27
28        intervals.append(
29            (process["pid"], start, end)
30        )
31
32        current_time = end
33
34    return intervals
35
36
37 # -----
```

2. SJF CODE

```
40
41 def sjf(processes):
42     # Make a copy so original data is preserved
43     remaining = processes.copy()
44
45     current_time = 0
46     intervals = []
47     while remaining:
48         # find processes that have arrived
49         ready = [
50             p for p in remaining
51             if p["arrival"] <= current_time
52         ]
53
54         # CPU idle if no process is available
55         if not ready:
56             next_arrival = min(
57                 p["arrival"] for p in remaining
58             )
59
60             intervals.append(
61                 ("IDLE", current_time, next_arrival)
62             )
63             current_time = next_arrival
64             continue
65
66         # Select shortest job
67         # Tie-breaking:
68         # Burst Time -> Arrival Time -> Process ID
69
70         process = min(
71             ready,
72             key=lambda p: (
73                 p["burst"],
74                 p["arrival"],
75                 p["pid"]
76             )
77         )
78
79         start = current_time
80         end = start + process["burst"]
81
82         intervals.append(
83             (process["pid"], start, end)
84         )
85
86         current_time = end
87         remaining.remove(process)
88
89     return intervals
90
91
92
93
94
95
```

3. FCFS AND SJF OUTPUT

```
manas_kumar_jha@LAPTOP-2LN5NJ4K:~/Operating-System-Lab$ python3 program2.py
FCFS AND SJF SCHEDULING
-----
Enter number of processes: 4
Enter Arrival Time for P1: 0
Enter Burst Time for P1: 7
Enter Arrival Time for P2: 2
Enter Burst Time for P2: 4
Enter Arrival Time for P3: 4
Enter Burst Time for P3: 1
Enter Arrival Time for P4: 5
Enter Burst Time for P4: 4

INPUT PROCESSES
-----


| PID | Arrival | Burst |
|-----|---------|-------|
| P1  | 0       | 7     |
| P2  | 2       | 4     |
| P3  | 4       | 1     |
| P4  | 5       | 4     |



=====
FCFS SCHEDULING
=====


| Process | Start | End |
|---------|-------|-----|
| P1      | 0     | 7   |
| P2      | 7     | 11  |
| P3      | 11    | 12  |
| P4      | 12    | 16  |



Sequence: P1 -> P2 -> P3 -> P4

=====
SJF SCHEDULING
=====


| Process | Start | End |
|---------|-------|-----|
| P1      | 0     | 7   |
| P3      | 7     | 8   |
| P2      | 8     | 12  |
| P4      | 12    | 16  |



Sequence: P1 -> P3 -> P2 -> P4
```

4. Input Validation

```
manas_kumar_jha@LAPTOP-2LN5NJ4K:~/Operating-System-Lab$ python3 program2.py
FCFS AND SJF SCHEDULING
-----
Enter number of processes: 2
Enter Arrival Time for P1: -1
Arrival Time cannot be negative.
Enter Arrival Time for P1: 0
Enter Burst Time for P1: 0
Burst Time must be greater than 0.
```

5. CPU IDLE Test

```
manas_kumar_jha@LAPTOP-2LNSN34K:~/Operating-System-Lab$ python3 program2.py
FCFS AND SJF SCHEDULING
-----
Enter number of processes: 4
Enter Arrival Time for P1: 2
Enter Burst Time for P1: 5
Enter Arrival Time for P2: 4
Enter Burst Time for P2: 3
Enter Arrival Time for P3: 5
Enter Burst Time for P3: 2
Enter Arrival Time for P4: 8
Enter Burst Time for P4: 4

INPUT PROCESSES
-----


| PID | Arrival | Burst |
|-----|---------|-------|
| P1  | 2       | 5     |
| P2  | 4       | 3     |
| P3  | 5       | 2     |
| P4  | 8       | 4     |


-----

FCFS SCHEDULING
-----


| Process | Start | End |
|---------|-------|-----|
| IDLE    | 0     | 2   |
| P1      | 2     | 7   |
| P2      | 7     | 10  |
| P3      | 10    | 12  |
| P4      | 12    | 16  |


Sequence: P1 -> P2 -> P3 -> P4

-----

SJF SCHEDULING
-----


| Process | Start | End |
|---------|-------|-----|
| IDLE    | 0     | 2   |
| P1      | 2     | 7   |
| P3      | 7     | 9   |
| P2      | 9     | 12  |
| P4      | 12    | 16  |


Sequence: P1 -> P3 -> P2 -> P4
```